

SECTION 16780

CCTV SYSTEM SPECIFICATION

PART 1 GENERAL

1.1 SUMMARY

- A. In general, the CCTV Contractor selected shall be responsible for providing a complete and operable CCTV system as described herein. This work shall include final connections to existing CCTV equipment and successful operation of the new equipment with the existing equipment. The CCTV Contractor shall provide a written response to each paragraph in this Specification to verify compliance.

1.2 RELATED DOCUMENTS

1.3 SUBMITTALS

- A. Before executing any work, submit for approval, 7 copies of information (in completely marked and coordinated package) to assure full compliance with the contract requirements. Information shall be submitted for all material and equipment the CCTV Contractor proposes to furnish to accomplish the contract work. Submittals for each manufactured item shall contain descriptive literature (equipment specification), equipment drawings, diagrams and catalog cuts of the proposed equipment. The submittal shall also include manufacturer names, catalog models and numbers, layout dimensions and capacities. Drawings shall indicate adequate clearance for operation, maintenance and replacement of operating equipment and devices.
- B. The following operational and maintenance material shall be submitted at the completion of the work:
 - 1). Three bound books, with index and index tabs, to include the following for all products and devices furnished under this Specification.
 - a). Hardware manuals
 - b). Maintenance manuals
- C. List of recommended spare equipment, along with quantities the Owner should maintain on site.
- D. Final approved set of all shop drawingsubmittals.

1.4 INSTALLATION AND START-UP REQUIREMENTS

- A. Micron will specify locations of all field devices prior to installation by the CCTV Contractor and will be shown on Engineering Design package documents.
 - B. Provide a written punch list of items remaining to be completed to Micron and/or General Contractor two (2) weeks prior to startup date.
- 1.5 Start-up and testing of the system will be done by the CCTV Contractor with the assistance of Micron. During start-up the CCTV Contractor will be responsible for repairing any mistakes in wiring or installation immediately. As a minimum, one CCTV technician familiar with the installation shall be available at all times during start-up for the repairs.
- 1.6 AS-BUILT DOCUMENTATION
- A. The Contractor shall keep one print of each drawing at their shop and in the field. These prints are to be marked-up as work progresses. The Contractor will provide as-built drawings at the end of this job. The as-built drawings shall be created using the marked-up prints as reference. A Mylar pencil plot of the floor plans with devices, equipment and notes only will be given to the CCTV Contractor. The CCTV Contractor shall be required to draw in the as-built conduit, wirefill, color coding and miscellaneous changes to create "as built" Mylar drawings from these.
- 1.7 WARRANTY
- A. The CCTV Contractor shall guarantee all materials and workmanship for a period of one year after acceptance of the building by the Owner. The CCTV Contractor shall make all necessary repairs, adjustments, and replacements at no cost to the Owner during this period. Provide a written letter of warranty to Micron on all equipment supplied by the CCTV Contractor outlining start-up dates, duration of warranty and terms of warranty. Refer to Section 01700 for additional requirements.

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

- A. The CCTV Contractor shall furnish all equipment required for a complete installation which is not supplied by the Owner or other contractors as part of the requirements listed in other sections of this specification. Where the CCTV Contractor is responsible for installing equipment furnished by the Owner or others, he will take full responsibility for the care and security of that equipment upon receipt of the equipment at the Job Site.
- B. All equipment furnished by the CCTV Contractor shall be kept secure and dry at the job site. Equipment that can be affected by ambient conditions shall be

stored in a conditioned environment. The CCTV Contractor shall provide all required storage facilities for this equipment on the jobsite.

- C. All CCTV Contractor furnished equipment shall be new and left in their original container until installed. The owner reserves the right to inspect this equipment prior to installation and reject any equipment not cared for in an appropriate or defined fashion.
- D. Refer to Section 16010 for additional requirements.

2.2 CCTV CAMERAS

A. Cameras :

- 1). Fixed: Small dome POE. Compatible with Verint VMS 7.5 and capable of at least 106°. Lux 0.25 color and 0.06 BW or IR illumination $\geq$ 2mp. H.264 compression. Environmental housing as needed
- 2). PTZ: VMS Compatible as mentioned in fixed category. Capable of 30X optical zoon. Outdoor ready.
 - a).

2.3 CONDUIT, RACEWAY AND FITTINGS

- A. Conduit: All interior conduits shall be EMT, size noted on Drawings as required by NEC cable fill. Exterior conduit shall be rigid steel. Conduit shall meet the following requirements.

2.4 All conduits shall be UL listed with approved fittings.

- A. Conduit shall be sized and supported in accordance with the NEC. Sizes shown on the drawings are to be used for bidding and should conform to the NEC; however, the Contractor shall be responsible for installing the correct size conduit. Tube fill in conduits shall be limited to 40% fill. Minimum size conduit is 3/4 inch,
 - 1). Running threads shall not be permitted and approved threaded couplings shall be used on full weight conduit. Each end of any conduit terminating in a box of any kind shall be provided with a lock-nut inside and outside of the box and with an approved insulating-type bushing.
 - 2). Conduit bends shall be the long radius type without kinks, flattening or crushing. For all bends in conduits 1 inch or larger standard elbows shall be used except where long pulls are involved. Exposed conduits shall be run at right angles or parallel to walls and floors indoors and to the supporting structures outdoors and all changes in direction shall be

made with conduit fittings or standard elbows. Conduit runs outdoors shall be installed with a slight incline in such a manner as to insure against trouble from the collection of trapped condensation.

- 3). For above grade conduit installations, as a minimum pull boxes shall be installed in all straight runs of conduit a minimum of every 100 feet; in all runs with one right angle bend a minimum of 75 feet; in all runs with two or three right angle bends a minimum of 50 feet. This requirement does not relieve the Contractor of the responsibility for providing pull points on all long conduit runs as required, so as not to over-stress and/or damage the wire or cable.
- 4). Open conduit ends, including those terminating in boxes, shall be plugged or capped as soon as installed, and so maintained during construction to prevent the entrance of moisture and dirt. All conduits shall be carefully cleaned and dried inside before the installation of wire.
- 5). All conduits installed outdoors or where subject to water, must enter equipment and junction boxes from the side of the bottom, where top entry is a necessity; watertight conduit hubs shall be used.
- 6). All runs of rigid steel conduit in excess of 100 feet without a right angle pull point. Shall be provided with an expansion joint.

B. Flexible Metal Conduit:

- 1). Flexible metal conduit shall be installed where called for in the drawings, to provide a flexible connection to equipment requiring position adjustments, or where fished through walls, above ceiling, or within other building cavities as shown on drawings or approved by the owner, flexible metal conduit shall not be used where EMT can be utilized.
- 2). Flexible metal conduit shall be UL approved with EEL approved angle wedge type insulated throat fittings specifically manufactured for use with flexible metal conduit. Flexible metal conduit shall be ½ inch minimum.

C. Liquid Tight Flexible Metal Conduit

- 1). Liquid-tight flexible metal conduit where called for shall be fabricated of galvanized steel flexible tubing, with a thermo-plastic jacket extruded over the tubing, Anaconda Seal-tite extra flexible. Liquid tight flexible metal conduit shall be the same size as the conduit system to which it connects unless noted otherwise.

- 2). Fittings for liquid tight flexible metal conduit shall be galvanized malleable iron.
- 3). Liquid tight flexible metal conduit shall be used for all flexible connections in the basement and penthouse and for all connections to instruments having a NEMA 3, 4, 7 or 9 enclosure.

D. Pull Boxes, Junction Boxes and Equipment:

- 1). Suitable boxes shall be provided where indicated on drawings, called for in the specifications, or as may be required to facilitate the installation and pulling of wires or cables.
- 2). Boxes shall be sized to accommodate wiring without sharp bends or overcrowding, and shall in all cases comply with NEC requirements and the cable manufacturer's requirement.
- 3). All boxes shall be independently and adequately supported and shall not be supported by the conduit system.

2.5 EQUIPMENT MOUNTING

- A. Provide all necessary hangers, brackets, steel frames, equipment racks, and pipe stands necessary to properly support all equipment and conduit as required. Where possible, parallel conduits shall be grouped neatly on structural racks and secured in position.
- B. Provide channel support to support panels, terminal boxes, and groups of instruments. Equipment shall not be mounted directly on walls or ceilings.
- C. All hangers, brackets and channel supports shall be galvanized steel, Uni-strut.
- D. All structural steel members or steel frames shall be hot-rolled carbon steel. Shapes shall conform with types already in use in the facility. Steel members shall be welded to existing structures in accordance with the American Welding Society for procedures, appearance and quality of welds.
- E. All steel shall be galvanized except steel members welded to existing steel which are presently painted and are used solely to support the new equipment or conduit.
- F. All fasteners, supports, clamps and anchors shall be of the correct type for the purpose. Fasteners shall be as follows:
 - 1). In hollow or lath construction; toggle or machine bolts, with suitable bearing plates.

- 2). For structural iron, machine bolts or approved clamps. Beam clamps can be used on structures free from vibration.
 - 3). For solid masonry, metallic expansion shields and machine screws. In concrete, provide approved inserts.
 - 4). Power driven fasteners and anchors may be used in areas designated by Micron. Where used, it will be the Contractor's responsibility to use the correct anchor and charge for the type of construction encountered.
- G. All brackets, hangers, bolts and nuts shall be galvanized.
- H. Twenty percent spare space shall be provided in all racks and each spare space shall be not less than 3 inches for future conduit.

2.6 WIRE AND CABLE

- A. All conductors shall be new, stranded copper of 98 percent conductivity and marked with identification in accordance with the regulations of Underwriters Laboratories.
- 1). Cable shall be as follows:

PART 3 EXECUTION

3.1 GENERAL

- A. The CCTV system shall include all devices, components, parts, wire, cable, conduit, etc. as specified herein, or in other Sections of the Specification, and connected to perform all functions and operate according to the system description and function contained in Part 1. The CCTV Contractor shall coordinate with other contractors working in the same areas or providing support services (120 volt AC) to his equipment.
- B. The CCTV Contractor shall:
- 1). Furnish and install all CCTV components along with all appurtenant parts, cables, raceway, mounting brackets, and supports required for a complete and operable system unless otherwise covered under another section.

3.2 EQUIPMENT INSTALLATION AND LABELING REQUIREMENTS

- A. Equipment shall be installed in accordance with the Drawings and Specifications and in strict accordance with the manufacturer's

recommendations. All equipment which must be serviced, operated or maintained shall be located in fully accessible locations. Locations shown on the drawings are diagrammatic but approximate the ideal location for the equipment at the time of design. Changes to these locations shall be coordinated with the Owner. Micron will assign all camera #'s.

- B. All electrical equipment must be located so as to provide minimum working space as required by the NEC.
- C. The CCTV Contractor shall coordinate with the Architect as to the exact location of all equipment and conduit locations that he can utilize.

3.3 DEVICE INSTALLATION AND LABELING REQUIREMENT

- A. All equipment and devices shall have nameplates. Nameplates shall be permanently adhered to the device with screws or double sided tape specifically made for this application. Nameplates shall include the device tag numbers and, if noted on the Drawings, the equipment description.

3.4 CONDUIT/WIRE INSTALLATION AND LABELING

- A. All wire and cable shall be continuous from terminal to device with no splices.
- B. All cables and wire shall have wire labels at each end and in junction boxes which shall be Brady Data type. Wire labels shall include tag numbers as listed on the interconnection diagram.
- C. All cables within the relay panels should be run in slotted plastic wire duct. Where wire duct is not provided, cables shall be installed using Thomas and Betts (T&B) nylon ty-wraps and anchor blocks.
- D. All control wiring shall be verified as to field device and cable integrity before being terminated at either end per the following procedure:
 - 1). Attach DVM across control cable at security panel end.
 - 2). Have person at device end short wires together: verify that continuity is established.
 - 3). Isolate wires at device and check for continuity from wire to shield, shield to ground, and wire to ground.
 - 4). If continuity is observed on any of the readings in step #3, correct problem or replace cable.
 - 5). If all readings in step #3 show, no continuity, then label and terminate cable at this time.

- 6). When terminating field devices, strip back shield, cut drain wire, and wrap any exposed shielding with black Scotch 33 vinyl tape. Terminate shield drain wire at control device.
- E. Wires shall be installed in strict accordance with the National Electrical Code, and in particular with Article 725. All communication, signal and instrument cable with insulation of less than 600 volts (Class 2 and 3) shall be kept separate by at least 2" from all power and Class 1 control circuits or a partition shall be installed between these classes of cables.
- F. Where adjacent terminal blocks are jumpered together as shown on the drawings, standard plug-in jumpers shall be provided and installed by the security contractor. Wire jumpers shall be used between fused terminal blocks and non-adjacent terminal blocks, and shall have one wire label in the middle where 6" or less, and a wire label on both ends where more than 6".
- G. Any penetration of a fire wall shall be sealed to maintain the fire rating of the wall.
- H. Pull boxes shall be 4 inches square by 2-1/8 inches deep for all installations except where 1-1/4 inch EMT is used. Where 1-1/4 inch EMT connects to a pull box, the box shall be 4-11/16 inches square by 2-1/8 inches deep. All boxes shall have a sheet metal cover plate and a nameplate identifying the number for cables in the box. Refer to the interconnection diagrams for tag numbers.
- I. All 120 volt power cables shall be installed in separate raceway or enclosures from low voltage cables.

END OF SECTION